

RODEO INFORMATION NETWORK

rinrodeo.com

Full Developer Specification

Data Entry · RPI · TPI · SRI · Scoring Rules · Phased Roadmap

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Section 1 — Glossary of Rodeo Terms

This glossary is provided so the developer understands rodeo-specific words used throughout this document. Please read this section before building any features.

Term	Definition
Cover / Covered	Roughstock only. The rider STAYED ON the animal for the full 8 seconds. This is a successful ride.
Did Not Cover (DNC)	Roughstock only. The rider was BUCKED OFF before 8 seconds ended. Score = 0. No official score is recorded.
No Time (NT)	Timed events only. The rider received NO official time. This happens when they knock a barrel, miss a loop, break a rule, or are disqualified. Time = NULL. Score = 0.
Qualified Ride	Same as 'Covered.' The rider completed the 8-second ride. This is required to receive a score.
Clean Run	Timed events only. The rider completed the run with NO penalties and NO disqualification.
Re-ride	When the bull or bronc does not perform well, the judges may OFFER the rider a chance to try again on a different animal. See Section 5 for full rules.
Flank Strap	A strap placed around the animal's flank to encourage bucking. If it falls off early, a re-ride may be offered.
Go-Round	One round of competition within an event. A rodeo may have multiple go-rounds (e.g., Round 1, Round 2, Short Round/Finals).
Short Round / Finals	The final round of a multi-go rodeo, where only the top-scoring riders compete.
Roughstock	Events where the rider tries to stay on a bucking animal: Bull Riding, Saddle Bronc Riding, Bareback Riding.
Timed Events	Events measured by time: Barrel Racing, Breakaway Roping, Tie-Down Roping, Steer Wrestling, Team Roping, Goat Tying, Pole Bending, Team Pinning.
Stock Contractor	A company or person who supplies the bulls and broncs for a rodeo event.
RPI	Rider Performance Index. A 0–100 score measuring how well a RIDER performs over time. Explained in Section 7.
TPI	Timed Performance Index. A 0–100 score measuring how well a HORSE performs in timed events over time. Explained in Section 8.
SRI	Stock Rating Index. A 0–100 score measuring how well a BULL or BRONC performs as a bucking animal. Explained in Section 9.
Header / Heeler	Team Roping only. The Header ropes the steer's horns first. The Heeler ropes the steer's back legs second. They are two separate riders.
Penalty	Extra time added to a timed event score for rule violations. Example: 5 seconds for knocking over a barrel in barrel racing.

NFR

National Finals Rodeo. The biggest professional rodeo championship held in Las Vegas in December.

Section 2 — Who Enters Data and When

User Type	What They Do
Rider (Athlete)	Enters their own profile information and self-reports their competition results. This is Phase 1.
Animal Owner	Enters their animal's profile information. Connects the animal to their own rider profile if they are also a competitor.
Stock Contractor	Enters their company profile and their animal roster. Animals they own auto-connect to event results.
Event Organizer	Phase 3 only — will enter event results directly. In Phase 1 and 2, riders self-report.
Platform Admin	Can edit or delete any record. Can approve disputed results. Can grant Verified Event Partner status.

Section 3 — Rider Data Entry (All Fields)

This section lists every field the rider enters when creating their profile. Fields marked **REQUIRED** must be filled in. Fields marked *Optional* can be left blank.

3.1 Profile Setup (One-Time Registration)

IMPORTANT:

These fields are entered **ONCE** when the rider first creates their account. They can edit them later from their dashboard.

3.1.1 Personal Information

Field Name	Instructions
First Name	REQUIRED. Rider's legal first name.
Last Name	REQUIRED. Rider's legal last name.
Nickname	Optional. Example: 'Cowboy Joe'
Date of Birth	REQUIRED. Used to verify identity when claiming a profile. Format: MM/DD/YYYY.
Gender	REQUIRED. Options: Male, Female, Non-binary, Prefer not to say.
City	REQUIRED. City where the rider lives.
State / Province	REQUIRED. State or province where the rider lives.
Country	REQUIRED. Default: United States.
Profile Photo	Optional. Rider uploads a headshot photo. Max file size: 5MB. Accepted formats: JPG, PNG.
Biography	Optional. A short paragraph the rider writes about themselves. Max 500 characters.

3.1.2 Competition Classification

Field Name	Instructions
Division	REQUIRED. Select ONE: Youth, High School, College, Amateur, Open/Pro, Senior.
Primary Event	REQUIRED. Select ONE main event they compete in most often. See full event list in Section 3.1.4.
Secondary Events	Optional. Select any additional events they also compete in.
Years Competing	Optional. How many total years they have competed in rodeo. Enter a number.

3.1.3 Association Memberships

Optional. The rider selects which rodeo associations they are a member of. They can select multiple.

Association	Full Name
PRCA	Professional Rodeo Cowboys Association
WPRA	Women's Professional Rodeo Association
NHSRA	National High School Rodeo Association
NIRA	National Intercollegiate Rodeo Association (College)
USTRC	United States Team Roping Championships
Other	Free text field — rider types the name of the association.

3.1.4 Social Media & Contact

Field Name	Instructions
Email Address	REQUIRED. Used for login and notifications. PRIVATE — never shown publicly.
Phone Number	Optional. PRIVATE — never shown publicly.
Instagram	Optional. Rider pastes their Instagram profile URL.
TikTok	Optional. Rider pastes their TikTok profile URL. IMPORTANT: This is the most-used platform for young rodeo athletes.
Facebook	Optional. Rider pastes their Facebook profile URL.
YouTube	Optional. Rider pastes their YouTube channel URL.
X (Twitter)	Optional. Rider pastes their X (formerly Twitter) profile URL.
Personal Website	Optional. Rider pastes their personal or sponsor website URL.
Video Highlights	Optional. Rider can paste up to 5 YouTube or Vimeo video links to display on their public profile.

3.2 Animal Claim (Connecting a Horse to Rider Profile)

A rider can connect a horse to their profile. This links the horse's performance data to the rider. For example, a barrel racer can claim her horse so both the rider and horse have linked performance histories.

NOTE:

This is a Phase 2 feature. In Phase 1, the animal_id is just entered as a text field on each result record. The full claim/link system comes in Phase 2.

Step	Description
Search Animal	Rider searches for their animal by name, registration number, or owner name.

Animal ID	If found in the system, rider selects the animal and it links to their profile.
Add New Animal	If not found, rider fills out the Animal Profile form (see Section 4).
Ownership Claim	Rider selects their relationship: Owner, Trainer, Lessee, Competitor (borrowing). This affects whether they can edit the animal profile.

Section 4 — Rider Result Entry: Roughstock Events

Roughstock events are: Bull Riding, Saddle Bronc Riding, Bareback Riding, Steer Riding, Junior Bull Riding.

The rider enters a result after they have competed. This is the most important form on the platform.

4.1 Step 1: Select the Event

Step	Description
Search for Event	REQUIRED. Rider types the event name and date to search if it already exists in the system.
Select Existing Event	If the event already exists, rider selects it from the search results. All event details auto-fill.
Add New Event	If the event does NOT exist, the rider clicks 'Add New Event' and fills in: Event Name, Date, City, State, Event Tier (Local/Regional/Major/Championship), Organization name (optional).

4.2 Step 2: Select Competition Details

Field	Instructions
Event Type	REQUIRED. The specific roughstock event. Options: Bull Riding, Saddle Bronc, Bareback, Steer Riding, Junior Bull Riding.
Division	REQUIRED. Which division the rider competed in at this event: Youth, High School, College, Amateur, Open/Pro, Senior.
Go-Round	REQUIRED. Which round of the event this result is for. Options: Round 1, Round 2, Round 3, Short Round/Finals, Average (if event uses cumulative scoring). Default: Round 1.
Animal ID / Name	Optional. The name or ID number of the bull or bronc they rode (or attempted to ride). Rider types this in. In Phase 2, this will link to the animal's profile.

4.3 Step 3: Enter the Score



ACTION REQUIRED:

THE MOST IMPORTANT FIELD IS THE FIRST ONE. The rider must first select whether they covered the animal or not. This controls which other fields appear on the form.

FIELD: Did You Cover? (Did You Complete the 8-Second Ride?)

Selection	What Happens in the Form
-----------	--------------------------

YES — I Covered	The rider STAYED ON for 8 full seconds. Score fields appear. Rider enters scores below.
NO — I Did Not Cover	The rider was BUCKED OFF before 8 seconds. All score fields are HIDDEN. Final Score is automatically set to 0. Qualified Ride = false. The system records this as a 0-score result.

⚠ IMPORTANT:

DO NOT show score fields when the rider selects 'No.' Hiding these fields prevents confusion and data entry errors.

Score Fields — ONLY visible when 'Did You Cover?' = YES

Field	Instructions
Judge 1 Score	REQUIRED if covered. A number between 0 and 25. This is the score given by Judge #1 for the RIDER's performance.
Judge 2 Score	REQUIRED if covered. A number between 0 and 25. This is the score given by Judge #2 for the RIDER's performance.
Animal Score	REQUIRED if covered. A number between 0 and 50. This is the combined score for the ANIMAL's bucking performance, judged by both judges together.
Final Score	AUTO-CALCULATED. Final Score = Judge 1 Score + Judge 2 Score + Animal Score. Maximum possible = 100. DO NOT let the rider type this. Calculate it automatically.

```
// Final score calculation
final_score = judge1_score + judge2_score + animal_score

// Validation rules:
// judge1_score must be between 0 and 25
// judge2_score must be between 0 and 25
// animal_score must be between 0 and 50
// final_score maximum = 100
// final_score minimum (if covered) = 0
// If NOT covered: final_score = 0, qualified_ride = false
```

4.4 Step 4: Re-ride

A re-ride is a special situation in roughstock events. See **Section 5** for the full re-ride explanation and database logic.

Field	Instructions
Was a Re-ride Offered?	YES or NO checkbox. Default = NO.
Did You Accept the Re-ride?	Only visible if re-ride was offered. YES or NO. Default = NO.

⚠ IMPORTANT:

If Re-ride Offered = YES and Re-ride Accepted = YES: A SECOND result entry form appears for the re-ride score. The re-ride score is what gets used for RPI. The original ride is still saved in the database but marked as replaced.

4.5 Step 5: Placement and Payout

Field	Instructions
Placement	Optional. Where the rider finished in their go-round. Enter a number: 1, 2, 3, etc. If they did not place in the money, leave blank.
Payout / Prize Money	Optional. PRIVATE — only visible to the rider on their private dashboard. How much money they won at this event. Enter in dollars.

📝 NOTE:

Placement affects the RPI placement bonus (+8 for 1st, +6 for 2nd, etc.). If left blank, no placement bonus is added.

Section 5 — Re-ride Rules (Critical for Developer)

This section is very important. Please read it carefully. A re-ride is a unique situation that requires special handling in the database and the RPI calculation.

5.1 What is a Re-ride?

✓ DEFINITION:

A re-ride is an offer given to a roughstock rider to try again on a DIFFERENT animal when the original animal did not perform adequately.

A re-ride may be offered when:

- The bull or bronc stops bucking before the 8 seconds are up
- The bull or bronc falls down
- The bull or bronc runs into the fence and stops bucking
- The flank strap falls off the animal
- The animal leaves the arena
- The animal does anything that prevents a fair ride opportunity

IMPORTANT: The JUDGES — not the rider — decide if a re-ride is offered. The rider cannot ask for a re-ride. The judges must offer it.

5.2 The Rider Has a Choice

Choice	What Happens
ACCEPT the re-ride	The rider chooses to try again on a DIFFERENT animal. If accepted, the new score (from the second ride) REPLACES the original score for RPI purposes. The original ride is still saved in the database but is NOT used in RPI calculations.
DECLINE the re-ride	The rider keeps their original score (even if it is 0 or very low). The original score is used in RPI calculations. This sometimes happens if the rider got a good score on a bad bull.

5.3 Database Logic for Re-rides

Store TWO separate records. Do NOT overwrite the original record.

Original Ride Record

```
result_id:      [unique ID]
rider_id:       [rider's ID]
```

```

event_id:           [event ID]
animal_id:          [first bull/bronc]
qualified_ride:     true or false
final_score:        [score or 0]
reride_offered:     true
reride_accepted:    true or false
reride_result_id:   [points to the re-ride record if accepted]
use_for_rpi:        false // Do NOT use this in RPI if re-ride was
accepted
                    true  // Use this in RPI if re-ride was DECLINED

```

Re-ride Record (only created if rider **ACCEPTED** the re-ride)

```

result_id:          [new unique ID]
rider_id:           [same rider's ID]
event_id:           [same event ID]
animal_id:          [DIFFERENT bull/bronc — the one used for the re-ride]
qualified_ride:     true or false
final_score:        [new score or 0]
is_reride:          true
original_result_id: [points back to the original ride record]
use_for_rpi:        true  // ALWAYS use the re-ride score for RPI

```

NOTE:

Summary: If a re-ride is accepted, use_for_rpi = false on the original and use_for_rpi = true on the re-ride. If declined, use_for_rpi = true on the original and no re-ride record is created.

5.4 What the Rider Sees in the Form

Here is the exact form logic for re-rides, written as a step-by-step process:

1. Rider enters their original ride result (covered or not covered, scores if covered).
2. Below the score fields, show a checkbox: "Was a re-ride offered by the judges?" — YES / NO.
3. **If NO:** Form is complete. Save the record with reride_offered = false, use_for_rpi = true.
4. **If YES:** Show a second checkbox: "Did you accept the re-ride?" — YES / NO.
5. **If YES, I accepted:** Show a NEW set of score fields (same as original: Did You Cover, Judge 1, Judge 2, Animal Score). This is the re-ride score. Save original record with use_for_rpi = false. Save re-ride record with use_for_rpi = true.
6. **If NO, I declined:** Form is complete. Save original record with reride_offered = true, reride_accepted = false, use_for_rpi = true.

Section 6 — Rider Result Entry: Timed Events

Timed events are: Barrel Racing, Pole Bending, Breakaway Roping, Tie-Down Roping, Steer Wrestling, Goat Tying, Team Roping, Team Pinning.

6.1 Step 1: Select the Event

Same as roughstock Section 4.1. Rider searches for the event or adds a new one.

6.2 Step 2: Select Competition Details

Field	Instructions
Event Type	REQUIRED. The specific timed event.
Division	REQUIRED. Same options as roughstock.
Go-Round	REQUIRED. Same options as roughstock.
Animal / Horse ID	Optional. The name or ID of the horse used. Links to TPI records in Phase 2.
Partner (Team Roping only)	REQUIRED for Team Roping. The rider selects whether they are the Header or the Heeler. They also enter their partner's name or RIN ID. See Section 6.5.

6.3 Step 3: Enter the Time



ACTION REQUIRED:

THE MOST IMPORTANT FIELD IS THE FIRST ONE. The rider must first select whether they received an official time or a No Time. This controls which other fields appear.

FIELD: Did You Receive an Official Time?

Selection	What Happens
YES — I received an official time	Time fields appear. Rider enters their run time and penalties.
NO — No Time / Did Not Score	All time fields are HIDDEN. Final Time = NULL. Clean Run = false. Score = 0 for RPI. Record this as no_time = true.



IMPORTANT:

'No Time' can happen for different reasons: missed a barrel, broke a barrier rope, missed a loop, or any disqualification. The rider does not need to explain why — just mark No Time.

Time Fields — ONLY visible when 'Did You Receive an Official Time?' = YES

Field	Instructions
-------	--------------

Raw Run Time	REQUIRED. The time the rider ran before penalties. Enter as seconds and hundredths. Example: 17.43
Number of Penalties	REQUIRED. How many penalties were assessed. Enter 0 if none. For barrel racing, each knocked barrel = 5-second penalty. For roping events, a broken barrier = 10-second penalty.
Penalty Seconds	AUTO-CALCULATED. Penalty Seconds = Number of Penalties × Penalty Value for that event. DO NOT let rider type this.
Final Time	AUTO-CALCULATED. Final Time = Raw Run Time + Penalty Seconds. This is the official time used for placement and RPI.
Clean Run	AUTO-CALCULATED. Clean Run = true if Number of Penalties = 0 AND No Time = false. Otherwise = false.

```
// Time calculation
penalty_seconds = number_of_penalties * penalty_value_per_event
final_time      = raw_run_time + penalty_seconds
clean_run       = (number_of_penalties == 0) AND (no_time == false)

// Penalty values by event type:
// Barrel Racing:    5 seconds per knocked barrel
// Pole Bending:    5 seconds per knocked pole
// Breakaway:       no penalty seconds — either clean or no-time
// Tie-Down Roping: 10 seconds barrier penalty; other infractions = no-time
// Steer Wrestling: 10 seconds barrier penalty
// Team Roping:     10 seconds barrier penalty (header's barrier)
// Goat Tying:      5 seconds per violation
```

6.4 Step 4: Placement and Payout

Same as roughstock Section 4.5.

6.5 Team Roping — Special Rules

⚠ IMPORTANT:

Team roping has TWO riders per run (Header and Heeler). Each rider gets their own separate result record. Both records are linked by a shared run_id.

Field	Instructions
Role	REQUIRED. Select: Header or Heeler.
Partner Name / RIN ID	REQUIRED. The name or RIN ID of the other roper (their partner). The system will attempt to match this to an existing rider profile.
Shared Run ID	AUTO-GENERATED by the system. A unique ID that links both the header's record and the heeler's record together. Neither rider types this in.

Completion Status

Options: Both loops clean (official time), Header clean / Heeler no-catch, Header no-catch, Barrier penalty (header). Each combination determines if a time is recorded.

```
// Database structure for team roping
// Two separate result records, linked by run_id

Record 1 (Header):
  result_id:    [unique ID 1]
  rider_id:     [header's RIN ID]
  run_id:       [shared UUID - same for both records]
  role:         'header'
  partner_id:   [heeler's RIN ID]
  final_time:   [same value as Record 2]
  placement:    [same value as Record 2]

Record 2 (Heeler):
  result_id:    [unique ID 2]
  rider_id:     [heeler's RIN ID]
  run_id:       [same shared UUID]
  role:         'heeler'
  partner_id:   [header's RIN ID]
  final_time:   [same value as Record 1]
  placement:    [same value as Record 1]

// Both riders' RPI values are updated independently from their own records
```

Section 7 — Animal Owner Data Entry (All Fields)

Animal profiles are created by owners, stock contractors, or riders who claim a horse. There are two types of animals: (1) Rough Stock — bulls and broncs, and (2) Performance Horses — horses used in timed events.

7.1 Fields for ALL Animals

Field	Instructions
Animal Name	REQUIRED. The registered name of the animal. Example: 'Bushwacker' or 'Lena's Dash For Cash'.
Animal Type	REQUIRED. Options: Bull, Saddle Bronc Horse, Bareback Horse, Barrel Horse, Breakaway Horse, Rope Horse, Steer Wrestling Horse, All-Around Horse, Other.
Breed	Optional. Example: Bucking bull breeds — Brahman cross, Brangus. Horse breeds — Quarter Horse, Paint, Appaloosa.
Color / Markings	Optional. Example: Red roan with blaze. Helps identify the animal.
Date of Birth / Age	Optional. Enter either the birth year or exact date.
Sex	Optional. Options: Bull, Steer, Gelding, Mare, Stallion.
Sire (Father)	Optional. Name of the animal's father. Important for breeding lineage records.
Dam (Mother)	Optional. Name of the animal's mother.
Registration Number	Optional. The animal's official registration number from a breed registry (AQHA, PtHA, etc.) or bucking bull registry (PRCA Animal Registry).
Photo	Optional. Photo of the animal. Max 5MB. JPG or PNG.
Video Links	Optional. Up to 5 YouTube or Vimeo links showing the animal competing.
Years Competing	Optional. How many years this animal has been used in competition.
Currently Active	REQUIRED. YES or NO. Is this animal currently competing? A retired or deceased animal should be set to NO.
Owner / Stock Contractor	REQUIRED. Link to the owner's rider profile or stock contractor profile.

7.2 Additional Fields for Rough Stock Only (Bulls and Broncs)

Field	Instructions
SRI (Stock Rating Index)	AUTO-GENERATED. Do not let the owner enter this. It is calculated by the system. See Section 9.
Buck-off Rate	AUTO-CALCULATED. Percentage of rides where the animal successfully bucked off the rider.
Average Animal Score	AUTO-CALCULATED. Average score the animal received from judges across all rides.

Total Rides	AUTO-CALCULATED. Total number of times this animal has been ridden at events recorded in the system.
--------------------	--

7.3 Additional Fields for Performance Horses Only

Field	Instructions
TPI (Timed Performance Index)	AUTO-GENERATED. Do not let the owner enter this. See Section 8.
Average Run Time	AUTO-CALCULATED. Average final time across all clean runs recorded for this horse.
Clean Run Rate	AUTO-CALCULATED. Percentage of runs where the horse ran clean with no penalties.
Total Runs	AUTO-CALCULATED. Total number of runs recorded for this horse.

Section 8 — RPI: Rider Performance Index Formula

The RPI is a number between 0 and 100 that measures how well a RIDER performs. A higher number is better. 50 is average. 80+ is excellent. 90+ is elite.

The RPI is NOT a ranking. Two riders can both have an RPI of 75.0 without being directly compared. Think of it like a golf handicap — it measures your performance, not your rank.

8.1 Overview of the Calculation Steps

Step	What Happens
Step 1	Calculate a raw event score for each competition result.
Step 2	Adjust the raw score based on the difficulty of the competition field.
Step 3	Apply a multiplier based on how big the event was (local vs championship).
Step 4	Apply a multiplier based on whether the result was self-reported or verified.
Step 5	Apply a time-decay weight — recent results count more than old results.
Step 6	Calculate the weighted average of all recent results.
Step 7	Apply a consistency bonus or penalty based on ride/run completion rate.
Step 8	Clamp the final number between 0 and 100.

8.2 Step 1a: Raw Event Score — Roughstock

```
// FIRST: Check if the rider covered
IF qualified_ride = false:
    base_performance_score = 0
    STOP — skip all other steps for this result

// If the rider covered:
base_performance_score = final_score
// (final_score is already 0-100 for roughstock)

// Add placement bonus:
IF placement = 1:    placement_bonus = 8
IF placement = 2:    placement_bonus = 6
IF placement = 3:    placement_bonus = 4
IF placement = 4:    placement_bonus = 3
IF placement = 5:    placement_bonus = 2
IF placement = 6-10: placement_bonus = 1
IF placement = blank or > 10: placement_bonus = 0

// Combine:
raw_event_score = base_performance_score + placement_bonus
// Clamp between 0 and 100:
```

```
raw_event_score = MIN(100, MAX(0, raw_event_score))
```

8.3 Step 1b: Raw Event Score — Timed Events

```
// FIRST: Check for no-time or disqualification
IF no_time = true OR disqualified = true OR clean_run = false:
    base_performance_score = 0
    STOP — skip all other steps for this result

// SAFETY CHECK: Prevent division by zero
IF (average_clean_time - winning_time) = 0:
    // All riders ran nearly the same time — give everyone 100
    base_performance_score = 100
ELSE:
    relative_ratio = (final_time - winning_time)
                    / (average_clean_time - winning_time)
    base_performance_score = 100 - (relative_ratio × 30)

// Clamp between 40 and 100 for clean runs:
base_performance_score = MIN(100, MAX(40, base_performance_score))

// Add placement bonus (same as roughstock above)
raw_event_score = base_performance_score + placement_bonus
raw_event_score = MIN(100, MAX(0, raw_event_score))

// NOTE: winning_time = fastest final_time in the same go-round and
division
// NOTE: average_clean_time = average of all clean final_times in the same
go-round
```

8.4 Step 2: Field Strength Multiplier

This step rewards riders who beat strong competition. Winning against elite riders earns more RPI than winning against beginners.

```
// Calculate average RPI of all riders in the same go-round and division
avg_field_rpi = average of rpi_current for all riders in this go-round

// If a rider has no RPI yet (provisional), use the division average
instead
// If fewer than 3 riders have an established RPI, skip this step:
IF count(established_riders_in_goround) < 3:
    field_strength_multiplier = 1.00
ELSE:
    field_strength_multiplier = 1.00 + ((avg_field_rpi - 50) / 100) × 0.20
    // Clamp between 0.85 and 1.20:
```

```

    field_strength_multiplier = MIN(1.20, MAX(0.85,
field_strength_multiplier))

// Examples:
// avg_field_rpi = 50 (average field) → multiplier = 1.00
// avg_field_rpi = 75 (strong field) → multiplier = 1.05
// avg_field_rpi = 25 (weak field) → multiplier = 0.95

```

8.5 Step 3: Event Tier Multiplier

```

IF event_tier = 'Local':           event_tier_multiplier = 1.00
IF event_tier = 'Regional':       event_tier_multiplier = 1.10
IF event_tier = 'Major':         event_tier_multiplier = 1.20
IF event_tier = 'Championship':  event_tier_multiplier = 1.30

```

8.6 Step 4: Verification Multiplier

```

IF result submitted by rider (self-reported):
verification_multiplier = 0.90
IF result submitted by Verified Event Organizer:
verification_multiplier = 1.00
IF result from official association API (Phase 4):
verification_multiplier = 1.05

```

8.7 Step 5: Calculate Full Event Performance Score

```

event_performance_score =
    raw_event_score
    × event_tier_multiplier
    × field_strength_multiplier
    × verification_multiplier

// Clamp between 0 and 100:
event_performance_score = MIN(100, MAX(0, event_performance_score))

```

8.8 Step 6: Recency Weight (Time Decay)

```

days_since_event = today's date - event date

IF days_since_event <= 90:    recency_weight = 1.00
IF days_since_event <= 180:  recency_weight = 0.75
IF days_since_event <= 365:  recency_weight = 0.50
IF days_since_event > 365:   EXCLUDE this result from rpi_current
calculation

                                (still counts for rpi_career and rpi_season)

```

8.9 Step 7: Weighted Average and Consistency

```
// For each result in the last 365 days:
weighted_score = event_performance_score × recency_weight

// Calculate the weighted average:
rpi_weighted_average = SUM(weighted_score) / SUM(recency_weight)

// Consistency adjustment:
// Roughstock: success_rate = qualified_ride_rate (rides covered / total
rides)
// Timed:      success_rate = clean_run_rate (clean runs / total runs)
consistency_adjustment = (success_rate - 0.50) × 10
consistency_adjustment = MIN(5, MAX(-5, consistency_adjustment))
// Example: 70% success rate → (0.70 - 0.50) × 10 = +2.0
// Example: 30% success rate → (0.30 - 0.50) × 10 = -2.0

// Final RPI:
rpi_current = rpi_weighted_average + consistency_adjustment
rpi_current = MIN(100, MAX(0, rpi_current))
```

8.10 RPI Status Rules

Status	Description
Provisional	0–4 verified results. RPI is shown but labeled 'Provisional.' Treat with caution.
Emerging	5–9 verified results. RPI is more reliable.
Established	10+ verified results. RPI is fully reliable.

8.11 Worked Example — Bull Riding

Here is a complete example showing how RPI is calculated for one result:

Input	Value
Rider	John Smith. Current RPI = 72.0.
Event	Major Rodeo, Regional tier
Result	Covered (8 seconds). Judge 1 = 22, Judge 2 = 21, Animal = 44. Final Score = 87.
Placement	2nd place → placement_bonus = 6
raw_event_score	87 + 6 = 93
Event tier	Regional → event_tier_multiplier = 1.10

Field strength	$\text{avg_field_rpi} = 68 \rightarrow \text{multiplier} = 1.00 + ((68-50)/100) \times 0.20 = 1.036$
Verification	Self-reported $\rightarrow \text{verification_multiplier} = 0.90$
event_performance_score	$93 \times 1.10 \times 1.036 \times 0.90 = 95.6 \rightarrow \text{clamped to } 95.6$
Recency	Event was 30 days ago $\rightarrow \text{recency_weight} = 1.00$
Consistency	Rider covered 6 of last 10 rides $\rightarrow \text{success_rate} = 60\% \rightarrow \text{adjustment} = +1.0$
Final rpi_current	Weighted average of all results + consistency adjustment

 NOTE:

The final rpi_current is calculated using ALL results in the last 365 days combined, not just this one result. This example shows the calculation for a single event_performance_score only.

8.12 Season and Career RPI

```
// rpi_season = average event_performance_score for results this calendar
year
// Season = January 1 of current year through today
// For High School / College riders, season is September 1 to August 31
// (configurable by division in a future update)

// rpi_career = average event_performance_score for ALL verified results,
all time
// No recency weighting - all results count equally
```

8.13 Recalculation Triggers

Recalculate rpi_current every time any of these events occur:

- A new result is added for this rider
- An existing result is edited
- A result is verified or un-verified
- The event tier is changed
- The rider changes division
- A re-ride result is accepted or declined

Section 9 — TPI: Timed Performance Index Formula

TPI is a number between 0 and 100 that measures how well a PERFORMANCE HORSE performs in timed events (barrel racing, roping, steer wrestling, etc.). A higher TPI = a better-performing horse.

TPI is calculated from the horse's competition history — specifically from the result records where this horse was used.

9.1 What Data is Used?

The TPI is built from result records that have this horse's animal_id. The system looks at every run where this horse competed and calculates how the horse performed relative to the field.

NOTE:

In Phase 1, animal_id is a text field entered by the rider. Full horse linking happens in Phase 2. TPI will be more accurate after Phase 2 when horses are properly claimed and linked.

9.2 TPI Calculation

```
// For each run where this horse competed:

// Step 1: Check if it was a clean run
IF no_time = true OR clean_run = false:
    horse_run_score = 0    // Did Not Score - counts against consistency
ELSE:
    // Step 2: Compare to field (same formula as rider RPI timed events)
    IF (average_clean_time - winning_time) = 0:
        horse_run_score = 100
    ELSE:
        relative_ratio = (final_time - winning_time)
                        / (average_clean_time - winning_time)
        horse_run_score = 100 - (relative_ratio * 30)
        horse_run_score = MIN(100, MAX(40, horse_run_score))

// Step 3: Apply recency weight (same as RPI)
horse_weighted_score = horse_run_score * recency_weight

// Step 4: Apply event tier multiplier (same as RPI)
horse_weighted_score = horse_weighted_score * event_tier_multiplier

// Step 5: Weighted average of all runs
tpi_weighted_avg = SUM(horse_weighted_score) / SUM(recency_weight)

// Step 6: Consistency bonus/penalty
horse_clean_rate = clean_runs / total_runs
horse_consistency = (horse_clean_rate - 0.50) * 10
horse_consistency = MIN(5, MAX(-5, horse_consistency))
```

```
// Step 7: Final TPI  
tpi_current = tpi_weighted_avg + horse_consistency  
tpi_current = MIN(100, MAX(0, tpi_current))
```

9.3 TPI Fields Stored in the Animal Table

Field	Description
tpi_current	Rolling weighted TPI based on recent runs (last 365 days).
tpi_season	Average run score within the current season.
tpi_career	Lifetime average run score across all records.
tpi_status	Provisional (0-4 runs), Emerging (5-9 runs), Established (10+ runs).
tpi_trend_90d	Change in TPI over last 90 days vs prior 90 days.
avg_time_clean_runs	Average final_time across all clean runs. Useful display stat.
clean_run_rate	Percentage of total runs that were clean (no penalties, no no-time).
total_runs	Total number of runs recorded for this horse.

Section 10 — SRI: Stock Rating Index Formula

SRI is a number between 0 and 100 that measures how good a BULL or BRONC is as a bucking animal. A higher SRI means the animal bucks harder, scores higher, and is harder to ride. A great bucking bull should have a HIGH SRI.

⚠ IMPORTANT:

IMPORTANT DIFFERENCE: For riders, high score is good because they rode well. For bulls, high SRI is also good — but for a DIFFERENT reason. A high SRI means the bull is hard to ride and scores high when a qualified ride happens. This makes the bull valuable.

10.1 What Makes a Good Bucking Animal?

Factor	Why It Matters
High Animal Score	When a rider covers the bull, the animal scored highly (out of 50). High animal scores = the bull bucks hard and moves unpredictably.
High Buck-off Rate	The bull bucks riders off frequently. A bull that riders rarely cover is considered elite. 70%+ buck-off rate = very good bucking bull.
Consistency	The bull performs the same way every time. A bull that sometimes bucks hard and sometimes stops is less valuable than one that always bucks hard.

10.2 SRI Calculation

```
// Gather all result records where this animal competed
// For each result:

// Component 1: Animal Score Component (out of 50 points)
// Only use results where the rider covered (qualified_ride = true)
// Because animal scores on uncovered rides are not always recorded
IF qualified_ride = true AND animal_score > 0:
    animal_score_pct = (animal_score / 50) × 100
    // Convert to 0-100 scale

// Component 2: Buck-off Rate Component
buckoff_rate = rides_not_covered / total_rides
buckoff_component = buckoff_rate × 100
// Example: 7 of 10 riders bucked off → buckoff_rate = 0.70 →
buckoff_component = 70

// Component 3: Weighted average of animal score (with recency weights)
FOR each covered_ride in last 365 days:
    weighted_animal_score = animal_score_pct × recency_weight ×
    event_tier_multiplier
```

```

avg_weighted_animal_score = SUM(weighted_animal_score) /
SUM(recency_weight_for_covered_rides)

// Component 4: Combine components
// 60% weight on average animal score, 40% weight on buck-off rate
sri_base = (avg_weighted_animal_score × 0.60) + (buckoff_component × 0.40)

// Consistency bonus (same concept as RPI)
// A consistent buckers gets a small bonus
score_variance = standard_deviation of animal_score_pct across all rides
IF score_variance < 8:   sri_consistency_bonus = +3   // Very consistent
IF score_variance < 15: sri_consistency_bonus = +1   // Moderately
consistent
IF score_variance >= 15: sri_consistency_bonus = -2   // Inconsistent

// Final SRI:
sri_current = sri_base + sri_consistency_bonus
sri_current = MIN(100, MAX(0, sri_current))

```

10.3 SRI Fields Stored in the Animal Table

Field	Description
sri_current	Rolling weighted SRI based on recent events (last 365 days).
sri_season	SRI based on results within the current season only.
sri_career	Lifetime average SRI across all recorded events.
sri_status	Provisional (0-4 events), Emerging (5-9), Established (10+).
buckoff_rate	Percentage of rides where the animal successfully bucked off the rider.
avg_animal_score	Average animal score (out of 50) across all covered rides.
total_rides	Total number of recorded rides on this animal.
sri_trend_90d	Change in SRI over last 90 days vs. prior 90 days.

Section 11 — Phased Roadmap

The platform is built in 4 phases. Animals, Stock Contractors, and Event Organizer profiles are already built. Phase 1 focuses on activating riders and getting scores into the system.

PHASE

1

MVP — Get Live and Get Scores

Riders register, enter results, RPI calculates. Everything else waits.



NOTE:

Goal: Get 100+ athletes actively using the platform and trusting the RPI number.

Feature	Description
Athlete Registration	New rider creates account, generates RIN ID. All profile fields from Section 3.1.
Profile Claim	Existing athlete records can be claimed by creating an account and verifying identity via email + date of birth match.
Score Self-Entry	Rider enters results using forms defined in Sections 4 and 6. Self-reported data.
RPI Live	RPI calculates automatically after each new result entry. Shows Provisional / Emerging / Established status.
Public Profile Page	Clean public page at rinrodeo.com/athletes/[rin-id]. Shows RPI, career stats, recent results, social links. Mobile-friendly.
Basic Shareable Link	Copy profile URL. One-tap share buttons: Facebook, Instagram, X (Twitter), TikTok.
QR Code (Private)	QR code on private dashboard only. For event check-in in Phase 2.
Basic Event Creation	Rider can create a new event record when logging a result if the event does not already exist.

PHASE

2

Platform Enrichment — Activate the Ecosystem

Dashboard, animals, social cards, NIL flag, event calendar.

Feature	Description
Private Dashboard	Full performance dashboard: RPI trend chart, last 5/10/season/career averages, performance by arena, by stock contractor, by region.
Animal Claim (Phase 2)	Rider searches for and claims their horse or bull. Connects the animal profile to their rider profile. TPI/SRI begin calculating.
Social Sharing Cards	Auto-generated graphics: season summary, RPI milestone, event win. One-tap post to Instagram, TikTok, Facebook.
NIL Flag	Rider toggles 'Open to Partnerships' on their private dashboard. Shows publicly on profile when enabled.

Sponsor Field	Public sponsor/brand affiliations field added to rider profile.
Event Organizer Reviews	Athletes leave star rating + written review of event organizers. Shown on organizer public profile.
Enhanced Event Calendar	Public event finder with filters: discipline, division, state, date, entry fee, prize pool, deadline.
Notifications	Email alerts: RPI change, result posted, upcoming event in your area, new event in your discipline.
TikTok Social Link	Add TikTok to all social media link fields (currently missing).
Confidence Score Display	Show rpi_confidence_score alongside the RPI number so users know how reliable it is.

PHASE

3

Event Organizer Direct Entry

Organizers post results — higher accuracy, verified data.

Feature	Description
Organizer Login Portal	Verified Event Partners can log in and submit results for their events directly.
Bulk Result Upload	Organizer uploads a CSV file with all results. System auto-matches RIN IDs by name and date of birth.
Result Verification Flow	When organizer posts a result, rider receives a notification to confirm or flag a discrepancy. Disputed results go to admin queue.
Verified Result Badge	Results from Verified Event Partners show a checkmark badge on the rider profile. Self-reported results show a different indicator.
Verified Partner Badge	Event organizers who meet quality standards receive a Verified Event Partner badge on their public profile.
Admin Dispute Queue	Internal admin tool to review and resolve disputed results.

PHASE

4

API & Scale — Automate and Monetize

Data feeds, NIL marketplace, mobile app, fantasy.

Feature	Description
Association API Feeds	Automated results from PRCA, WPRA, NHSRA. Requires formal partnership agreements. Results marked as association-verified.
NIL Brand Marketplace	Brand/sponsor portal for western lifestyle companies to browse athletes by discipline, region, performance tier, and NIL status.
College NIL Compliance	NIL deal documentation tool for college athletes, NCAA-compliant.
Mobile App	iOS and Android. QR event check-in (built in Phase 1) fully activates. Push notifications. Score entry from phone.

Fantasy Rodeo	Season-long fantasy competition tied to NFR. Fan engagement and platform traffic driver.
Outbound Data API	Public API for rodeo media, broadcasters, stats sites. Tiered access model.

Section 12 — Developer Checklist: Critical Rules

This section is a summary of the most important rules for the developer to remember. Please review this list before finalizing any feature.

12.1 Score Entry Rules — NEVER Forget These

ACTION REQUIRED:

If 'Did Not Cover' is selected in roughstock: ALL score fields must be HIDDEN. Set final_score = 0 and qualified_ride = false automatically. Do not let the rider enter a score.

ACTION REQUIRED:

If 'No Time' is selected in timed events: ALL time fields must be HIDDEN. Set final_time = NULL and clean_run = false automatically. Do not let the rider enter a time.

ACTION REQUIRED:

Final Score (roughstock) must be AUTO-CALCULATED. Do not let the rider type the final score. Calculate it from Judge 1 + Judge 2 + Animal Score.

ACTION REQUIRED:

Final Time (timed events) must be AUTO-CALCULATED. Do not let the rider type the final time. Calculate it from Raw Time + Penalty Seconds.

12.2 Re-ride Rules — NEVER Forget These

IMPORTANT:

If re-ride accepted: Set use_for_rpi = false on the ORIGINAL record. Set use_for_rpi = true on the RE-RIDE record. NEVER use both in the same RPI calculation.

IMPORTANT:

If re-ride declined: Set use_for_rpi = true on the ORIGINAL record. Do not create a re-ride record.

12.3 RPI/TPI/SRI Formula Rules

Rule	What to Do
Division by zero	In the timed event formula: if (average_clean_time - winning_time) = 0, set base_performance_score = 100. Do not let the formula divide by zero.
Team roping	Create TWO result records (one for header, one for heeler). Link them with a shared run_id. Update each rider's RPI separately.

use_for_rpi field	Only include results where use_for_rpi = true in RPI calculations. Skip all others.
365-day cutoff	Results older than 365 days are excluded from rpi_current. They still count for rpi_career.
Clamp all values	rpi_current, tpi_current, and sri_current must always be between 0 and 100. Use MIN(100, MAX(0, value)) at the end of every calculation.
Recalculate on change	Recalculate RPI/TPI/SRI any time a result is added, edited, or its verification status changes.

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Questions? Contact the project lead before building any feature in this document.